

Iman Gholami

+1 (240) 960 6899 | igholami.com | imangholami77@gmail.com | igholami@umd.edu
[linkedin.com/in/imangholami](https://www.linkedin.com/in/imangholami) | github.com/igholami | [Google Scholar](https://scholar.google.com/citations?user=igholami)

RESEARCH INTERESTS

Approximation Algorithms, Network Design, Combinatorial Optimization, Algorithmic Game Theory, Metric Distortion, Formal Verification, AI for Mathematics

EDUCATION

University of Maryland, College Park

MD, USA

PhD Candidate in Computer Science

Jan 2023 – Present

- GPA: 3.97

Sharif University of Technology

Tehran, Iran

BSc in Computer Engineering

Sep 2017 – May 2022

- GPA: 3.57

EXPERIENCE

Research Assistant

Jan 2023 – Present

University of Maryland

College Park, MD

- Research on network design problems, including variations of classic connectivity problems, for better approximation factors than state-of-the-art algorithms.
- DARPA expMath team research towards verifiable mathematical proving through Artificial Intelligence (AI).
- Research on distortion problems, specifically classic problems with access to ordinal information instead of cardinal information.
- DARPA QuICC team research on Quantum-Inspired solvers for complex optimization problems.
- Academic mentoring of incoming PhD students and coordination of research teamwork.
- External reviewer for major conferences including FOCS, STOC, SODA, SPAA, and CDC.

Teaching Assistant

Jan 2023 – Present

University of Maryland, College Park

MD, USA

- Computational Genomics (Rob Patro, Spring 2026)
- Introduction to Computational Game Theory (MohammadTaghi Hajiaghayi, Fall 2025)
- Computer and Network Security (Michael Marsh, Fall 2024)
- Introduction to Computational Game Theory (MohammadTaghi Hajiaghayi, Fall 2023)
- Algorithms (Max Morowski & Clyde Kruskal, Spring 2023)

Software Engineer Intern

Jun 2026 – Aug 2026

Axiom

New York, NY

- Advancing formal verification of Zero-Knowledge Virtual Machines (ZK VMs) in Lean 4.
- Designed and implemented a Reinforcement Learning (RL) evaluation system to optimize Large Language Models (LLMs) and reasoning models for automated proof generation.

Software Engineer Intern

May 2025 – Aug 2025

Axal (YC W25)

San Francisco, CA

- Design and implement code analysis tools for information extraction using Artificial Intelligence (AI), with the goal of rewriting legacy code into modern, maintainable systems.

Software Engineer and Team Leader

Nov 2018 – Dec 2022

Balad (Local Navigation and Maps App, 30M+ Installs)

Tehran, Iran

Teaching Assistant

Sep 2019 – Jun 2022

Sharif University of Technology

Tehran, Iran

- Theory of Machines and Languages (Ali Movaghar, Fall 2021)
- Design Algorithm (Hamid Zarrabi-Zadeh, Fall 2021)
- Modern Information Retrieval (Mahdiah Soleymani Baghshah, Spring 2021)
- Discrete Structures (Hamid Zarrabi-Zadeh, Spring 2021)
- Design Algorithm (Mohammad Ali Abam, Fall 2019)

Scientific and Executive Committee Member

Jan 2020 – Aug 2023

Iran's National Olympiad in Informatics (INOI)

Tehran, Iran

Co-Founder and Software Engineer <i>Analysaur Team (Multi-Platform Smart Advertising Systems)</i>	Jun 2019 – Feb 2020 <i>Tehran, Iran</i>
Software Engineer Intern <i>Bazaar (Local Application Marketplace, 100M+ Installs)</i>	Jul 2018 – Oct 2018 <i>Tehran, Iran</i>
Technical Head of Infrastructure Team <i>Sharif Artificial Intelligence Challenge Yearly Recurring Event</i>	Feb to Mar 2018,19,20 <i>Tehran, Iran</i>

HONORS/AWARDS

Best Paper Award FOCS 2025	Dec 2025
2nd Place in ICPC NAS, Mid-Atlantic USA Region	Mar 2023
Bronze Medal in National Olympiad of Computer Engineering	Nov 2021
4th Place in ACM/ICPC Asia Region, Tehran Site	Dec 2020
2nd Place in ACM/ICPC Asia Region, Tehran Site	Dec 2017
Silver Medal in International Olympiad in Informatics (IOI)	Aug 2017
Gold Medal in Iran National Olympiad in Informatics (INOI)	Aug 2016
Silver Medal in Iran National Olympiad in Informatics (INOI)	Aug 2015

PUBLICATIONS

Beyond the Library: An Agentic Framework for Autoformalizing Research Mathematics <i>arXiv preprint, 2026</i>
Quiet Planting for k-SAT, Multiple Solutions of Arbitrary Geometry <i>Conference on Learning Theory (COLT 2026)</i>
Bi-Criteria Metric Distortion <i>International Conference on Learning Representations (ICLR 2026)</i>
Breaking a Long-Standing Barrier: $2\text{-}\epsilon$ Approximation for Steiner Forest <i>IEEE Symposium on Foundations of Computer Science (FOCS 2025)</i>
2-Approximation for Prize-Collecting Steiner Forest <i>Journal of the ACM (JACM 2025), ACM-SIAM Symposium on Discrete Algorithms (SODA 2024)</i>
Prize-Collecting Forest with Submodular Penalties: Improved Approximation <i>Integer Programming and Combinatorial Optimization (IPCO 2025)</i>
Prize-Collecting Steiner Tree: A 1.79 Approximation <i>ACM Symposium on Theory of Computing (STOC 2024)</i>

SKILLS

Research & Leadership: Technical Leadership, Teamwork, Communication, Conflict Resolution, Academic Mentoring, Research Strategy, Cross-functional Collaboration
Theoretical Computer Science (TCS): Approximation Algorithms, Network Design, Algorithmic Game Theory, Combinatorial Optimization, Graph Theory, Data Structures, Problem Solving
AI & Machine Learning: Large Language Models (LLMs), Automated Theorem Proving, Generative AI, AI-driven Analysis, Applied ML
Software Engineering: Distributed Systems, System Architecture, Static Analysis, Large-Scale Backend Systems, High-Performance Computing, API Design
Languages: C++ (200K+ LOC), Python (200K+ LOC), Lean 4, Kotlin, TypeScript, Go, Java, Bash, SQL
Frameworks & Libraries: React, FastAPI, Django, Node.js, NumPy, Pandas, Scikit-learn
Cloud, DB & Tools: AWS, Docker, Kubernetes, PostgreSQL, MongoDB, Redis, Git, CI/CD, Linux, Nginx

REFERENCES

Prof. Hajiaghayi, MohammadTaghi <i>Minker Professor at Computer Science Department, University of Maryland, College Park</i>	hajiagha@umd.edu
Dr. Chakraborty, Diptarka <i>Assistant Professor at School of Computing, National University of Singapore</i>	diptarka@comp.nus.edu.sg
Dr. Abam, Mohammad Ali <i>Associate Professor at Computer Engineering Department, Sharif University of Technology</i>	abam@sharif.edu
Dr. Zarrabi-Zadeh, Hamid <i>Associate Professor at Computer Engineering Department, Sharif University of Technology</i>	zarrabi@sharif.edu